Congenital Disorder of Glycosylation Type IIw (SLC37A4-CDG): A Comprehensive Disease Characterization Report

Disease: Congenital disorder of glycosylation type IIw (SLC37A4-CDG) **MONDO ID:** MONDO:0030437 · **OMIM:** 619525 · **Category:** Mendelian (autosomal dominant) **Gene:** *SLC37A4* (glucose-6-phosphate transporter, G6PT / G6PT1) · **HGNC:** 4061

Summary

SLC37A4-CDG (also designated congenital disorder of glycosylation type IIw, CDG-IIw; OMIM 619525; MONDO:0030437) is an **ultra-rare, autosomal-dominant Type II congenital disorder of glycosylation** caused by a **single recurrent, mostly *de novo* nonsense variant** in *SLC37A4*: **c.1267C>T (p.Arg423*)**. This is a striking example of one gene producing two mechanistically opposite diseases. Biallelic **loss-of-function** of *SLC37A4* causes the classical autosomal-recessive **glycogen storage disease type 1b (GSD1b)**. In contrast, the heterozygous p.Arg423* truncation removes the C-terminal endoplasmic reticulum (ER)-retention signal of the glucose-6-phosphate transporter (G6PT) while leaving the transporter's catalytic transport function largely intact. The truncated protein is **mislocalized from the ER to the Golgi apparatus**, where its presence disrupts Golgi glycosylation homeostasis and produces a congenital disorder of glycosylation rather than a glycogen storage disease.

Clinically, SLC37A4-CDG presents in infancy or childhood with **hepatopathy (liver dysfunction)** and a **severe, liver-derived multifactorial coagulopathy** — characteristically combining **antithrombin deficiency with factor XI deficiency** (and less often protein C, protein S, or factor IX deficiency), a profile that is distinct from the coagulation derangements of liver failure, disseminated intravascular coagulation (DIC), or vitamin K deficiency. Both bleeding and thrombotic complications can occur. Variable **cardiac involvement** and other multisystem features are also reported. Biochemically, patients show a **Type II serum**

transferrin/N-glycan pattern, abnormal serum N-glycoprotein glycoforms, abnormal **bikunin proteoglycan isoforms**, and **endoglycosidase-H-sensitive serum N-glycans**, reflecting immature Golgi N-glycan processing and combined N-/O-glycosylation plus proteoglycan defects.

Diagnosis rests on recognition of the Type II CDG biochemical signature confirmed by **molecular identification of the recurrent c.1267C>T (p.Arg423*) variant**, typically via exome or genome sequencing. There is **no curative or disease-specific therapy**; management is **supportive**, centered on correcting the coagulopathy (plasma/factor replacement, careful peri-procedural hemostatic monitoring) and surveillance of liver and cardiac function. Because the disorder is defined by a small number of published cases (roughly a dozen individuals across the founding reports), much of the epidemiology, natural history, penetrance, and prognosis remains incompletely characterized.

Key Findings

Finding 1 — A single recurrent dominant variant (p.Arg423*) mislocalizes G6PT to the Golgi and causes CDG-IIw

SLC37A4-CDG is caused by a **recurrent heterozygous, predominantly *de novo* variant, c.1267C>T (p.Arg423*)** in *SLC37A4*. Three independent reports converge on this identical variant. Marquardt et al. (2020) established the molecular mechanism: the nonsense mutation truncates the C-terminal cytoplasmic tail of the glucose-6-phosphate transporter, abolishing its ER-retention signal and creating a weak Golgi-retention signal, so the transporter is intracellularly **mislocalized to the Golgi**. As the authors state, *"The mutation abolishes the ER retention signal of the transporter and generates a weak Golgi retention signal. Intracellular mislocalization of the transporter leads to a congenital disorder of glycosylation instead of glycogen storage disease"* [PMID: 32884905].

Ng et al. (2021) reported a cohort of **seven heterozygous individuals** carrying the same variant, defining the core clinical triad: *"We report seven individuals who presented with liver dysfunction multifactorial coagulation deficiency and cardiac issues and were heterozygous for the same variant, c.1267C>T (p.Arg423*)"* [PMID: 33964207]. Wilson et al. (2021) independently described a second patient with the identical heterozygous *de novo* c.1267C>T (p.R423*) substitution [PMID: 33728255], reinforcing the recurrent, mutation-specific nature of the disorder.

This is the defining genetic feature of the disease: rather than a spectrum of loss-of-function alleles (as in recessive GSD1b), CDG-IIw is essentially a **single-variant, gain-of-mislocalization dominant disorder**.

Finding 2 — The disorder produces a Type II CDG serum pattern with combined N-/O-glycosylation and proteoglycan defects and a liver-derived coagulopathy

Raynor et al. (2021) provided detailed biochemical characterization of **six affected individuals**, demonstrating that the disorder involves multiple, simultaneous glycosylation abnormalities arising from disrupted Golgi homeostasis. They found *"abnormal patterns for various serum N-glycoproteins and bikunin proteoglycan isoforms, together with specific alterations of the mass spectra of endoglycosidase H-released serum N-glycans"* [PMID: 34245688]. The endoglycosidase-H sensitivity of serum N-glycans indicates **incomplete/ immature Golgi processing** of N-glycans — a hallmark of a Type II (Golgi-level) CDG.

Critically, the same work established the two central disease-level features: SLC37A4-CDG is *"characterized by a dominant inheritance and a major coagulopathy originating from the liver"* [PMID: 34245688]. The combination of abnormal N-glycoprotein glycoforms, altered O-linked proteoglycan (bikunin) isoforms, and a profound liver-derived hemostatic disturbance distinguishes this from other CDGs and from primary hepatic disease. Ng et al. (2021) corroborated the coagulation and hepatic phenotype in their seven-patient heterozygous cohort [PMID: 33964207].

Finding 3 — SLC37A4 encodes the ER glucose-6-phosphate transporter; its normal biology explains why a C-terminal truncation causes CDG rather than GSD

SLC37A4 encodes a **429-amino-acid, ~10-transmembrane-helix ER membrane protein** (G6PT) that transports glucose-6-phosphate (G6P) from the cytoplasm into the ER lumen, where glucose-6-phosphatase hydrolyzes it — a reaction central to glucose homeostasis and gluconeogenesis/glycogenolysis. Xia et al. (2025) solved cryo-EM structures of human G6PT (apo, G6P-bound, and chlorogenic-acid-bound), defining a substrate pocket with **subsite A (binding the phosphate)** and **subsite B (binding the glucose moiety)**, and showed that G6PT transport activity is enhanced by co-expression of glucose-6-phosphatase (G6PC) although the two proteins do not form a stable complex. As they summarize, *"The human glucose-6-phosphate transporter (G6PT) moves glucose-6-phosphate (G6P) into the lumen of endoplasmic reticulum, playing a vital role in glucose homeostasis"* [PMID: 41136424].

The key to understanding CDG-IIw lies in the transporter's C-terminus. Chen et al. (2000) showed that the C-terminal cytoplasmic tail contributes to folding/stability but that its most distal residues are dispensable for transport function: *"amino acids 415-417 in the cytoplasmic tail of the carboxyl-domain, extending from helix 10, also play a critical role in the correct folding of the transporter. However, the last 12 amino acids of the cytoplasmic tail play no essential role(s) in functional integrity"* [PMID: 10940311]. Because p.Arg423* truncates only the terminal residues, the mutant transporter **retains its transport capability but loses the C-terminal ER-targeting/retention signal** — precisely the combination that produces mislocalization to the Golgi (Finding 1) rather than a non-functional transporter. This mechanistic contrast explains why the dominant CDG is biochemically opposite to the recessive GSD1b caused by biallelic loss-of-function.

Finding 4 — Diagnosis combines a Type II serum transferrin/N-glycan signature with molecular confirmation; treatment is supportive

Diagnostic biochemistry integrates several assays: abnormal serum N-glycoprotein glycoforms (a **Type II CDG transferrin pattern**), abnormal bikunin proteoglycan isoforms, and endoglycosidase-H-sensitive serum N-glycans reflecting immature Golgi processing [PMID: 34245688; 33728255; 32884905]. Raynor et al. explicitly note the diagnostic utility of these markers: *"these data complement previous findings, help to better delineate SLC37A4-CDG and could present interest in diagnosing this disease"* [PMID: 34245688]. Definitive diagnosis requires **molecular confirmation of the recurrent heterozygous de novo c.1267C>T (p.Arg423*)** variant, generally identified via exome/genome sequencing or targeted testing.

Unlike GSD1b, the dominant CDG is distinguished by a **liver-derived coagulopathy** as its dominant feature, rather than the fasting hypoglycemia and chronic neutropenia that dominate GSD1b. **No disease-specific or curative therapy exists**; management is supportive and centers on correcting the coagulopathy (plasma or factor replacement as clinically indicated) and monitoring liver and cardiac function. As therapeutic context, sugar-substrate approaches have been trialed in the related *SLC37A4* disorder GSD1b — *"we hypothesized the same pathomechanism in GSD-Ib and started a therapeutic trial with oral galactose and uridine"* [PMID: 28126686] — but such approaches have not been established for CDG-IIw.

Finding 5 — The CDG coagulopathy has a characteristic factor profile that distinguishes it from liver failure, DIC, and vitamin K deficiency

Pascreau et al. (2023), in an ISTH state-of-the-art review of hemostatic defects in CDGs, defined the characteristic coagulation signature that applies to SLC37A4-CDG's liver-derived coagulopathy. CDG patients *"often present coagulation abnormalities characterized by low levels of procoagulant or anticoagulant factors. Antithrombin deficiency is frequently associated with factor XI deficiency and less frequently with a protein C, protein S, or factor IX deficiency. This coagulation profile differs from those observed in liver failure, disseminated intravascular coagulation, and vitamin K deficiency"* [PMID: 37193126]. This distinctive combination — **antithrombin** + **factor XI deficiency** as the core, with variable protein C/S and factor IX involvement — is diagnostically valuable because it separates a glycosylation-based coagulopathy from more common acquired causes.

Importantly, the hemostatic balance in CDG is precarious and bidirectional: *"Coagulopathy can lead to thrombotic and/or hemorrhagic complications"* [PMID: 37193126]. Patients are therefore at risk of **both bleeding and thrombosis**, requiring close hemostatic monitoring especially during acute illness, surgery, or other physiologic stress.

Mechanistic Model / Interpretation

Ordered causal chain

```
1. De novo point mutation SLC37A4 c.1267C>T  (germline, heterozygous)
       │  results in
       ▼
2. Premature stop codon p.Arg423* → truncation of the C-terminal
   cytoplasmic tail of the glucose-6-phosphate transporter (G6PT)
       │  which
       ▼
3. Removes the C-terminal ER-retention signal AND creates a weak
   Golgi-retention signal  (transport function itself preserved,
   because the distal ~12 residues are dispensable for transport)
       │  leads to
       ▼
4. Intracellular MISLOCALIZATION of G6PT from the ER to the Golgi
   apparatus  (dominant, gain-of-mislocalization effect)
       │  results in
       ▼
5. Disruption of Golgi glycosylation homeostasis
       │
       ├─► 5a. Impaired Golgi N-glycan maturation
       │        → endoglycosidase-H-sensitive serum N-glycans;
       │          Type II serum transferrin pattern
       │
       ├─► 5b. Abnormal O-linked proteoglycan processing
       │        → abnormal bikunin isoforms
       │
       └─► 5c. Under-/aberrant glycosylation of hepatically
                synthesized glycoproteins (incl. coagulation factors
                and their regulators)
       │  leads to
       ▼
6. Multisystem clinical manifestations:
     • Hepatopathy (liver dysfunction)
     • Liver-derived multifactorial coagulopathy
       (antithrombin ↓ + factor XI ↓ ± protein C/S ↓, factor IX ↓)
       → bleeding AND/OR thrombosis risk
     • Variable cardiac involvement
     • Other variable multisystem features
```

Steps 1–4 are directly demonstrated (mutation identification and cell-biological localization studies). Step 5 is supported by serum glycomics/proteoglycan data. The precise molecular link between Golgi-mislocalized G6PT and specific downstream glycosylation-enzyme dysfunction (step 5 → 6) is **partly inferred**: exactly how a mislocalized sugar-phosphate transporter perturbs Golgi glycosyltransferase/nucleotide-sugar homeostasis has not been fully resolved and represents the principal open mechanistic question.

One gene, two opposite diseases

Feature	GSD1b (recessive)	SLC37A4-CDG / CDG-IIw (dominant)
Gene	*SLC37A4*	*SLC37A4*
Inheritance	Autosomal recessive	Autosomal dominant (mostly *de novo*)
Molecular defect	Biallelic loss-of-function	Recurrent p.Arg423* truncation
Effect on transporter	Non-functional / absent transport	Transport preserved, ER-retention lost
Localization	ER (or degraded)	**Mislocalized to Golgi**
Disease class	Glycogen storage disease	Congenital disorder of glycosylation (Type II)
Hallmark features	Fasting hypoglycemia, hepatomegaly, neutropenia	Hepatopathy, liver-derived coagulopathy, cardiac issues
Glycosylation	Normal	Abnormal (Type II serum transferrin, EndoH-sensitive N-glycans, abnormal bikunin)

Ontology term suggestions

- **Disease:** MONDO:0030437 (SLC37A4-CDG / CDG-IIw)
- **Gene/Protein (GO / cellular component):** GO:0005783 endoplasmic reticulum; GO:0005794 Golgi apparatus; GO:0016021 integral component of membrane; GO:0015152 glucose-6-phosphate transmembrane transporter activity; GO:0015760 glucose-6-phosphate transport
- **Biological process:** GO:0006486 protein glycosylation; GO:0006487 protein N-linked glycosylation; GO:0018242 protein O-linking; GO:0006094 gluconeogenesis; GO:0042593 glucose homeostasis
- **Cell types (CL):** CL:0000182 hepatocyte (primary site of coagulation-factor synthesis); CL:0000091 Kupffer cell (context)
- **Anatomy (UBERON):** UBERON:0002107 liver; UBERON:0000948 heart; UBERON:0001981 blood vessel (thrombosis/bleeding)

- **Chemical entities (CHEBI):** CHEBI:4170 D-glucose 6-phosphate; CHEBI:17234 glucose; CHEBI:16709 1,5-anhydro-D-glucitol (1,5-AG; relevant to related G6PT neutropenia biology)

Section-by-Section Characterization

1. Disease Information

SLC37A4-CDG is a Mendelian, autosomal-dominant Type II congenital disorder of glycosylation caused by mislocalization of the ER glucose-6-phosphate transporter to the Golgi. **Identifiers:** MONDO:0030437; OMIM 619525; synonyms include **CDG-IIw**, **congenital disorder of glycosylation type IIw**, **SLC37A4-CDG**. It is distinct from GSD1b (also *SLC37A4*, OMIM 232220). Information is derived from **aggregated, disease-level case reports and small case series** (Marquardt 2020; Ng 2021; Wilson 2021; Raynor 2021), not from EHR/population resources.

2. Etiology

Primary cause: monogenic — the recurrent heterozygous *SLC37A4* c.1267C>T (p.Arg423*) variant [PMID: 32884905; 33964207; 33728255]. **Genetic risk factor:** this specific variant is the sole known cause; no susceptibility loci or modifier genes are established. Most cases are *de novo*, so an affected parent is not required. **Environmental risk/protective factors and gene-environment interactions:** none established; the disorder is not known to depend on environmental triggers. No protective alleles are known.

3. Phenotypes

Reported phenotypes (from small cohorts) with suggested HPO terms:

Phenotype	Type	Onset	Frequency (small cohorts)	HPO suggestion
Liver dysfunction / hepatopathy	Lab + clinical	Infancy–childhood	Core / most patients	HP:0001392 Abnormality of the liver; HP:0002910 Elevated hepatic transaminase
Multifactorial coagulopathy	Lab abnormality	Infancy–childhood	Core / most patients	HP:0001928 Abnormality of coagulation
Antithrombin deficiency	Lab abnormality	Variable	Characteristic	HP:0032154 Reduced antithrombin III activity
Factor XI deficiency	Lab abnormality	Variable	Frequent	HP:0004866 Abnormal factor XI
Bleeding and/or thrombosis	Clinical sign	Variable/ episodic	Variable	HP:0001892 Abnormal bleeding; HP:0001977 Abnormal thrombosis
Cardiac involvement	Clinical sign	Variable	Reported in Ng cohort	HP:0001627 Abnormal heart morphology
Abnormal serum transferrin glycosylation	Lab abnormality	Congenital	Diagnostic	HP:0003642 Abnormal isoelectric focusing of serum transferrin

Severity is **variable**; the coagulopathy can be severe and life-threatening. Quality-of-life data specific to CDG-IIw are not available given the tiny cohort.

4. Genetic / Molecular Information

Causal gene: *SLC37A4* (HGNC:4061), OMIM gene 602671, chromosome 11q23.3. **Pathogenic variant:** c.1267C>T, p.Arg423* — a **nonsense/truncating** variant; recurrent and mostly *de novo*; classified pathogenic. **Functional consequence:** not a simple loss of function but a **dominant gain-of-mislocalization** (ER→Golgi) with preserved transport activity [PMID: 32884905; 10940311]. Allele frequency: effectively absent from population databases (private/ de novo). **Modifier genes / epigenetics / chromosomal abnormalities:** none established for CDG-IIw. (Note: for the related GSD1b/neutropenia biology, heterozygous *SLC5A10*/SGLT5 variants modify 1,5-AG handling [PMID: 35506446; 37594549], but this is not established as a modifier of CDG-IIw.)

5. Environmental Information

No environmental, lifestyle, or infectious contributors are known. The disease is purely genetic.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the ordered causal chain. In brief: mutation → C-terminal truncation → loss of ER-retention + weak Golgi-retention → Golgi mislocalization of G6PT → disrupted Golgi glycosylation homeostasis → Type II N-glycan, O-glycan/proteoglycan, and coagulation-factor glycosylation defects → hepatopathy, coagulopathy, cardiac and multisystem manifestations. Molecular profiling to date is **glycomic/proteomic** (serum N-glycans, bikunin isoforms) [PMID: 34245688]; no transcriptomic, single-cell, or CRISPR-screen data specific to CDG-IIw are published.

7. Anatomical Structures Affected

Primary organ: liver (UBERON:0002107) — the source of the coagulopathy and site of hepatopathy. **Secondary/variable:** heart (UBERON:0000948); vasculature (thrombosis/hemorrhage). **Body systems:** hepatobiliary, hematologic/coagulation, cardiovascular. **Subcellular compartments:** endoplasmic reticulum (GO:0005783) and **Golgi apparatus (GO:0005794)** — the compartment where the mislocalized transporter exerts its pathogenic effect. **Cell type:** hepatocyte (CL:0000182). **Lateralization:** not applicable (systemic/metabolic disorder).

8. Temporal Development

Onset: congenital/pediatric — clinical presentation in infancy to childhood [PMID: 33964207]. **Course:** chronic, lifelong; the coagulopathy is a persistent risk that can fluctuate/episodically worsen with illness or procedures. Progression rate and long-term natural history are not well defined given limited follow-up. **Critical periods:** peri-procedural/peri-surgical and acute-illness windows are periods of heightened hemostatic vulnerability.

9. Inheritance and Population

Inheritance: autosomal dominant, predominantly **de novo** [PMID: 32884905; 33964207; 33728255]. **Penetrance/expressivity:** appears highly penetrant for the specific variant but with **variable expressivity**; formal estimates are lacking. **Epidemiology:** ultra-rare — only

~10–14 individuals reported across the founding papers; true prevalence/incidence unknown. No founder effect, consanguinity role (not applicable for dominant de novo), sex bias, or ethnic predilection has been established. **Carrier frequency:** not applicable (dominant, de novo).

10. Diagnostics

Biochemical screening: Type II serum transferrin isoform pattern; abnormal serum N-glycoprotein glycoforms; abnormal bikunin proteoglycan isoforms; **endoglycosidase-H-sensitive serum N-glycans** [PMID: 34245688]. **Coagulation testing:** the characteristic **antithrombin** + **factor XI deficiency** profile (± protein C/S, factor IX) helps distinguish it from liver failure, DIC, and vitamin K deficiency [PMID: 37193126]. **Molecular confirmation:** identification of the recurrent heterozygous *SLC37A4* c.1267C>T (p.Arg423*) variant — best via **whole-exome or whole-genome sequencing**, or targeted single-variant/single-gene testing. **Differential diagnosis:** GSD1b (recessive, same gene — but hypoglycemia/neutropenia dominant), other Type II CDGs, primary hepatic coagulopathy, DIC, vitamin K deficiency.

11. Outcome / Prognosis

No formal survival, mortality, or quality-of-life statistics exist for this ultra-rare disorder. The dominant prognostic concern is the **liver-derived coagulopathy**, which carries risk of both **hemorrhagic and thrombotic complications** [PMID: 37193126], alongside hepatopathy and variable cardiac involvement. Prognosis appears variable; close hemostatic and hepatic monitoring is the mainstay of risk reduction.

12. Treatment

No curative or disease-specific therapy. Management is **supportive**: correction of the coagulopathy with plasma and/or specific factor/anticoagulant-factor replacement guided by the individual factor profile, careful peri-procedural hemostatic planning, and surveillance/support of liver and cardiac function. Antithrombin concentrate and factor replacement may be considered for the specific deficiencies; anticoagulation decisions must balance the bidirectional bleeding/thrombosis risk. Sugar-substrate therapy (e.g., oral galactose ± uridine) has been trialed in the related disorder GSD1b [PMID: 28126686] but is **not established** for CDG-IIw. **Suggested NCIT terms:** NCIT:C15311 Supportive Care; NCIT:C561 Fresh Frozen Plasma; NCIT:C1685 Coagulation Factor.

13. Prevention

No primary prevention exists (mostly de novo). **Secondary prevention** = early biochemical/molecular diagnosis to guide hemostatic management and prevent bleeding/thrombotic events. **Tertiary prevention** = peri-procedural coagulation-factor optimization and organ-function surveillance. **Genetic counseling** is appropriate: recurrence risk for parents of a de novo case is low, but an affected individual would have a 50% transmission risk. **Prenatal/preimplantation testing** is technically feasible once the familial variant is known.

14. Other Species / Natural Disease

No naturally occurring animal disease attributable to this specific dominant *Slc37a4* mislocalization variant is reported. **Orthologs:** mouse *Slc37a4*, rat *Slc37a4* (well-conserved). Existing *Slc37a4* animal models pertain to the recessive GSD1b/neutropenia biology, not CDG-IIw. No zoonotic or cross-species relevance.

15. Model Organisms

No dedicated CDG-IIw animal model is published to date. Mechanistic work has relied on **in vitro cell-based localization studies** demonstrating ER→Golgi mislocalization of the mutant transporter [PMID: 32884905] and on **structural biology** (cryo-EM of human G6PT) [PMID: 41136424]. A knock-in mouse carrying the equivalent of p.Arg423* would be the logical model to recapitulate the dominant CDG phenotype and is a key gap.

Evidence Base

PMID	Title (abbrev.)	Type	Contribution
32884905	*SLC37A4-CDG: Mislocalization of G6PT to the Golgi*	Human + in vitro	Defines the founding mechanism: p.Arg423* abolishes ER-retention, creates weak Golgi-retention → mislocalization → CDG not GSD
33964207	*A mutation in SLC37A4 causes a dominantly inherited CDG*	Human (n=7)	Establishes recurrent variant + core triad: liver dysfunction, coagulopathy, cardiac issues
33728255	*SLC37A4-CDG: Second patient*	Human case	Independent confirmation of recurrent de novo c.1267C>T (p.R423*)
34245688	*SLC37A4-CDG: New biochemical insights … major coagulopathy*	Human (n=6)	Type II serum N-glycan / bikunin / EndoH signature; defines dominant inheritance + liver-derived coagulopathy; diagnostic markers
41136424	*Structural basis for transport/inhibition of G6PT*	Structural (cryo-EM)	Defines normal G6PT function, G6P binding subsites, G6PC interaction
10940311	*Structural requirements for stability/transport of G6PT*	In vitro	Shows C-terminal distal residues dispensable for transport → explains why p.Arg423* retains function while losing ER targeting
37193126	*Hemostatic defects in CDGs*	Review (ISTH)	Defines characteristic AT + FXI deficiency profile distinguishing CDG coagulopathy from liver failure/DIC/vitamin-K deficiency; bidirectional risk
28126686	*Oral galactose therapy in GSD1b*	Human trial	Therapeutic context for related SLC37A4 disorder (sugar-substrate approach)

Supporting/background literature on G6PT biology and GSD1: 10598822, 10518030, 25804016, 12192101, 11560776, 11121425, 10712583, and the SGLT5/1,5-AG neutropenia biology in related G6PT deficiency 35506446, 37594549.

Limitations and Knowledge Gaps

1. **Tiny evidence base.** The disease is defined by roughly a dozen published individuals, all carrying the same variant. Prevalence, incidence, penetrance estimates, sex ratio, natural history, and long-term prognosis are essentially unquantified.
2. **Mechanistic gap at step 5.** How a Golgi-mislocalized glucose-6-phosphate transporter specifically perturbs Golgi glycosylation enzymes / nucleotide-sugar or ion homeostasis is inferred, not mechanistically resolved. The link between the transporter's mislocalization and the observed N-/O-glycan and proteoglycan defects needs direct experimental demonstration.
3. **No animal model.** All mechanistic data are in vitro/structural; there is no in vivo model recapitulating the dominant CDG phenotype.
4. **Genotype restricted to one variant.** It is unknown whether other C-terminal truncating or ER-retention-disrupting variants could produce the same dominant CDG; the phenotype is currently variant-specific.
5. **No therapy data.** Management is empirically supportive; no trials, treatment-response rates, or targeted therapeutics exist for CDG-IIw specifically.
6. **Quality-of-life and outcome instruments** have not been applied to this cohort.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse (or zebrafish) model** carrying the murine equivalent of p.Arg423* to test whether Golgi mislocalization reproduces the hepatopathy, Type II glycosylation defect, and coagulopathy in vivo.
2. **Direct Golgi mechanistic studies:** in hepatocyte-like cells (e.g., patient iPSC-derived hepatocytes or organoids), map how mislocalized G6PT alters Golgi nucleotide-sugar/ion pools and glycosyltransferase activity — closing the step-5 inference gap.
3. **Systematic coagulation-factor phenotyping** across all reported and future patients to quantify the frequency of antithrombin, factor XI, protein C/S, and factor IX deficiencies and correlate with bleeding/thrombosis events, informing prophylaxis algorithms.
4. **Establish an international patient registry** to capture natural history, penetrance, cardiac phenotype spectrum, and long-term outcomes.

5. **Structure-guided variant survey:** use the cryo-EM G6PT structure [PMID: 41136424] and the C-terminal targeting data [PMID: 10940311] to predict and functionally test whether additional C-terminal variants can cause CDG-IIw, refining the genotype definition.
6. **Test candidate supportive/targeted therapies** (e.g., glycosylation-substrate supplementation, factor-specific replacement strategies) in the model system before clinical translation.

Report compiled from the confirmed knowledge state of a 5-iteration autonomous investigation. Evidence source types: human clinical case series (Marquardt, Ng, Wilson, Raynor), in vitro cell biology and biochemistry, structural biology (cryo-EM), and a clinical review of CDG hemostasis. Where data are absent for this ultra-rare disorder, this has been stated explicitly.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery